

Titration report



Method

Method ID	CDC_GranTED-DYN	Release state	Released
Name	Titration with HCl - Gran Verification	Modified by	PredefinedUser
Modified on	2/5/2026 6:46:26 AM		

Summary

Samples

Sample ID	Name	Result	State
1	0.1 M TRIS - 5 mL in 60 mL H2O	Sample size	5 mL
		Consumption R1	5.019684 mL
		Concentration R2	0.10039 mol/L
2	0.1 M TRIS - 5 mL in 60 mL H2O	Sample size	5 mL
		Consumption R1	5.001255 mL
		Concentration R2	0.10003 mol/L
3	0.1 M TRIS - 5 mL in 60 mL H2O	Sample size	5 mL
		Consumption R1	5.009554 mL
		Concentration R2	0.10019 mol/L

Statistics

R1

Mean	5.010164 mL	Minimum	5.001255 mL
Standard deviation	0.009230 mL	Rel. standard deviation	0.184 %
Total samples	3	Samples excluded	0

R2

Mean	0.10020 mol/L	Minimum	0.10003 mol/L
Standard deviation	0.00018 mol/L	Rel. standard deviation	0.180 %
Total samples	3	Samples excluded	0

Main Results

Sample ID1	Sample ID2	Result Title	Name	Result	State
0.1 M TRIS - 5 mL in 60 mL H2O		Consumption	R1	5.019684 mL	OK
0.1 M TRIS - 5 mL in 60 mL H2O		Concentration	R2	0.10039 mol/L	OK
0.1 M TRIS - 5 mL in 60 mL H2O		Consumption	R1	5.001255 mL	OK
0.1 M TRIS - 5 mL in 60 mL H2O		Concentration	R2	0.10003 mol/L	OK
0.1 M TRIS - 5 mL in 60 mL H2O		Consumption	R1	5.009554 mL	OK

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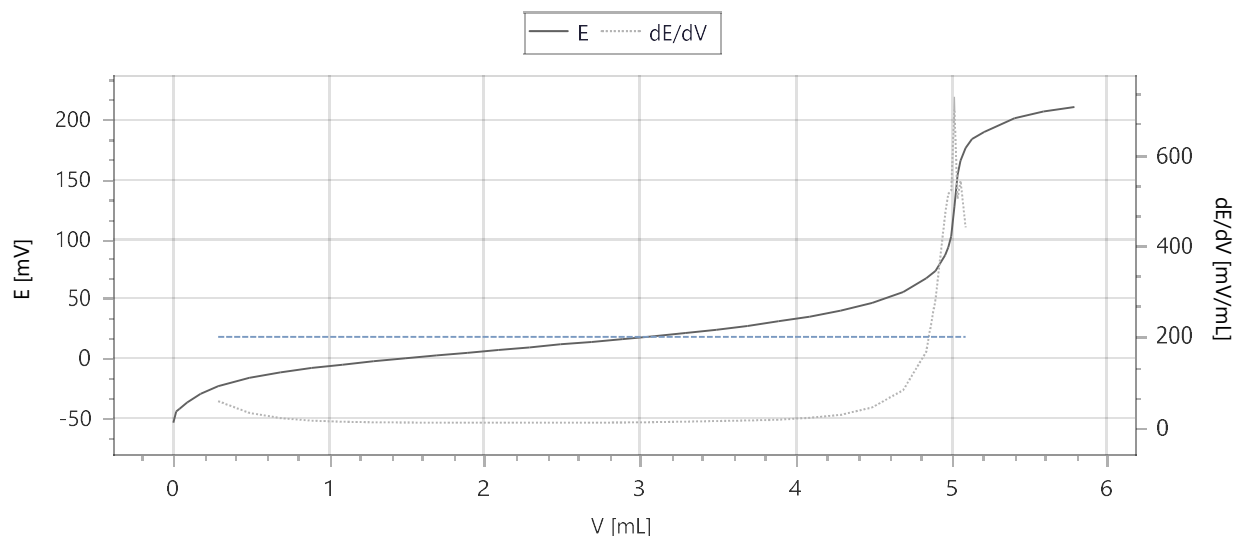
0.1 M TRIS - 5 mL
in 60 mL H₂O

Concentration R2

0.10019 mol/L OK

Titration (EQP) [1] Scope 1/1, Sample 1/3

E - V & dE/dV - V Curve Scope 1/1, Sample 1/3



Measured Values Scope 1/1, Sample 1/3

Time [s]	V [mL]	Meas. value [mV]	dV [mL]	Temp. [°C]	dE [mV]	dE/dV [mV/mL]	EQP
0	0.0000	-54.0		25.0			
33	0.0200	-44.5	0.0200	25.0	9.5		
36	0.0400	-42.3	0.0200	25.0	2.2		
40	0.0900	-37.1	0.0500	25.0	5.2		
43	0.1760	-29.8	0.0860	25.0	7.3		
48	0.2885	-23.4	0.1125	25.0	6.4	58.68	
54	0.4885	-16.4	0.2000	25.0	7.0	32.92	
60	0.6885	-11.8	0.2000	25.0	4.6	21.34	
66	0.8885	-8.1	0.2000	25.0	3.7	15.80	
71	1.0885	-5.3	0.2000	25.0	2.8	13.41	
76	1.2885	-2.6	0.2000	25.0	2.7	12.23	
81	1.4885	0.0	0.2000	25.0	2.6	11.73	
86	1.6885	2.4	0.2000	25.0	2.4	11.55	
91	1.8885	4.5	0.2000	25.0	2.1	11.42	
96	2.0885	6.8	0.2000	25.0	2.3	11.22	
100	2.2885	9.1	0.2000	25.0	2.3	11.26	
103	2.4885	11.6	0.2000	25.0	2.5	11.37	
108	2.6885	13.6	0.2000	25.0	2.0	11.54	
112	2.8885	16.0	0.2000	25.0	2.4	11.75	
117	3.0885	18.5	0.2000	25.0	2.5	12.52	
122	3.2885	21.0	0.2000	25.0	2.5	13.62	
125	3.4885	23.9	0.2000	25.0	2.9	14.98	
130	3.6885	27.1	0.2000	25.0	3.2	16.22	
133	3.8885	30.9	0.2000	25.0	3.8	17.95	
137	4.0885	34.9	0.2000	25.0	4.0	22.17	

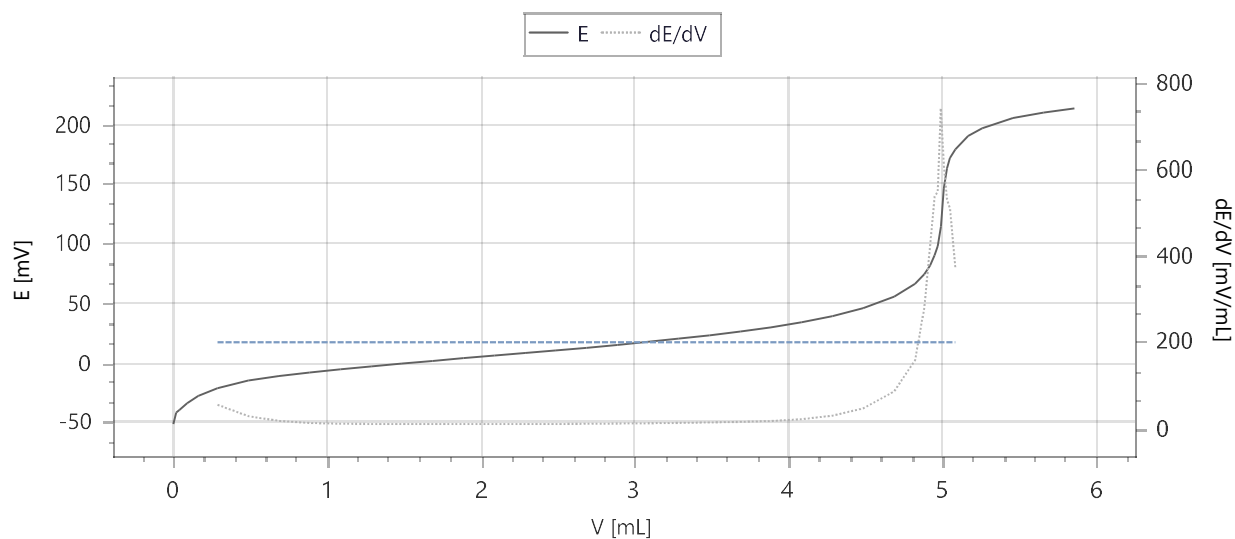
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142	4.2885	39.8	0.2000	25.0	4.9	28.65	
146	4.4885	46.2	0.2000	25.0	6.4	44.81	
150	4.6885	55.5	0.2000	25.0	9.3	82.80	
157	4.8360	67.0	0.1475	25.0	11.5	166.67	
160	4.8970	73.2	0.0610	25.0	6.2	283.95	
172	4.9575	86.1	0.0605	25.0	12.9	467.28	
180	4.9775	93.0	0.0200	25.0	6.9	513.67	
183	4.9975	101.9	0.0200	25.0	8.9	521.14	
198	5.0175	126.8	0.0200	25.0	24.9	705.01	
	5.019684	129.7				705.27	EQP
208	5.0375	153.6	0.0200	25.0	26.8	502.85	
212	5.0575	165.2	0.0200	25.0	11.6	542.80	
215	5.0895	175.9	0.0320	25.0	10.7	440.42	
218	5.1305	183.5	0.0410	25.0	7.6		
228	5.2090	189.5	0.0785	25.0	6.0		
232	5.4050	200.8	0.1960	25.0	11.3		
235	5.5890	206.5	0.1840	25.0	5.7		
238	5.7890	210.4	0.2000	25.0	3.9		

Titration (EQP) [1] Scope 1/1, Sample 2/3

E - V & dE/dV - V Curve Scope 1/1, Sample 2/3



Measured Values Scope 1/1, Sample 2/3

Time [s]	V [mL]	Meas. value [mV]	dV [mL]	Temp. [°C]	dE [mV]	dE/dV [mV/mL]	EQP
0	0.0000	-51.6		25.0			
33	0.0200	-42.0	0.0200	25.0	9.6		
36	0.0400	-39.7	0.0200	25.0	2.3		
41	0.0900	-34.0	0.0500	25.0	5.7		
45	0.1625	-27.9	0.0725	25.0	6.1		
50	0.2880	-21.5	0.1255	25.0	6.4	55.56	
56	0.4880	-15.1	0.2000	25.0	6.4	29.32	
61	0.6880	-11.3	0.2000	25.0	3.8	18.31	
68	0.8880	-8.3	0.2000	25.0	3.0	13.55	

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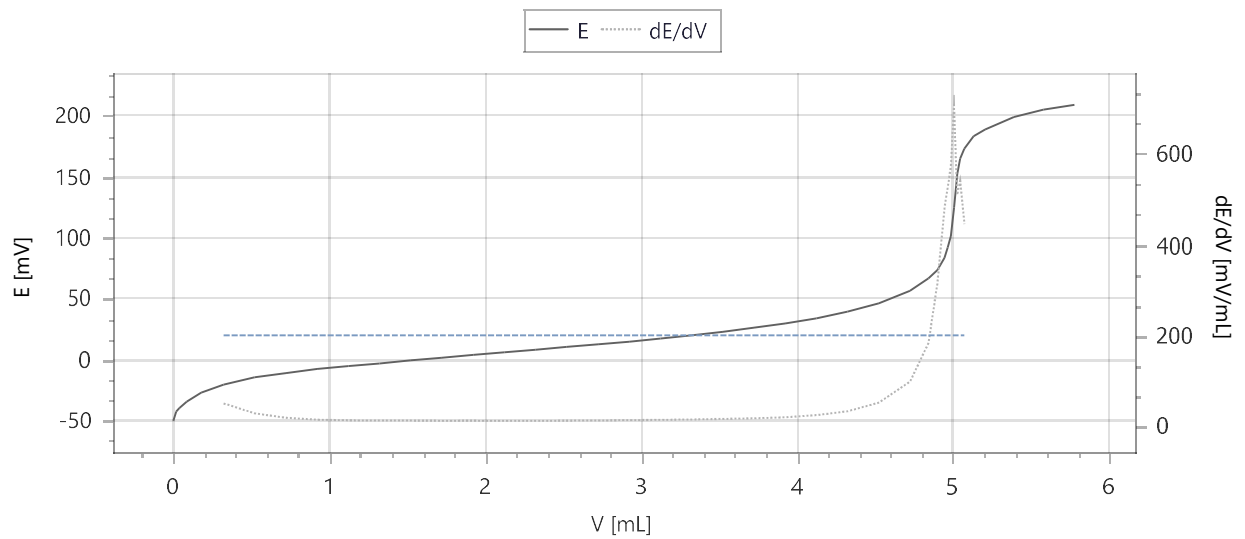
74	1.0880	-5.7	0.2000	25.0	2.6	11.71	
79	1.2880	-3.2	0.2000	25.0	2.5	11.22	
85	1.4880	-0.9	0.2000	25.0	2.3	11.26	
90	1.6880	1.4	0.2000	25.0	2.3	11.09	
95	1.8880	3.6	0.2000	25.0	2.2	10.93	
100	2.0880	5.8	0.2000	25.0	2.2	10.85	
105	2.2880	7.9	0.2000	25.0	2.1	10.93	
110	2.4880	10.2	0.2000	25.0	2.3	11.15	
114	2.6880	12.4	0.2000	25.0	2.2	11.60	
119	2.8880	14.8	0.2000	25.0	2.4	12.17	
123	3.0880	17.3	0.2000	25.0	2.5	12.80	
128	3.2880	20.0	0.2000	25.0	2.7	13.65	
133	3.4880	22.9	0.2000	25.0	2.9	14.76	
137	3.6880	26.0	0.2000	25.0	3.1	16.04	
142	3.8880	29.5	0.2000	25.0	3.5	18.05	
146	4.0880	33.8	0.2000	25.0	4.3	22.27	
150	4.2880	38.8	0.2000	25.0	5.0	30.46	
155	4.4880	45.5	0.2000	25.0	6.7	47.34	
161	4.6880	55.2	0.2000	25.0	9.7	86.21	
167	4.8250	66.1	0.1370	25.0	10.9	158.69	
174	4.8850	73.9	0.0600	25.0	7.8	280.58	
181	4.9230	81.2	0.0380	25.0	7.3	425.61	
189	4.9520	89.8	0.0290	25.0	8.6	537.13	
192	4.9720	97.6	0.0200	25.0	7.8	547.96	
203	4.9920	114.1	0.0200	25.0	16.5	740.84	
	5.001255	128.9				742.93	EQP
212	5.0120	146.1	0.0200	25.0	32.0	607.15	
217	5.0320	162.7	0.0200	25.0	16.6	531.13	
220	5.0520	171.5	0.0200	25.0	8.8	509.14	
228	5.0870	178.6	0.0350	25.0	7.1	369.71	
232	5.1700	189.8	0.0830	25.0	11.2		
235	5.2615	196.3	0.0915	25.0	6.5		
238	5.4615	204.9	0.2000	25.0	8.6		
241	5.6615	209.6	0.2000	25.0	4.7		
244	5.8615	213.1	0.2000	25.0	3.5		

Titration (EQP) [1] Scope 1/1, Sample 3/3

E - V & dE/dV - V Curve Scope 1/1, Sample 3/3

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Measured Values Scope 1/1, Sample 3/3

Time [s]	V [mL]	Meas. value [mV]	dV [mL]	Temp. [°C]	dE [mV]	dE/dV [mV/mL]	EQP
0	0.0000	-50.3		25.0			
34	0.0200	-42.6	0.0200	25.0	7.7		
37	0.0400	-39.7	0.0200	25.0	2.9		
41	0.0880	-34.4	0.0480	25.0	5.3		
46	0.1795	-27.2	0.0915	25.0	7.2		
53	0.3230	-20.6	0.1435	25.0	6.6	49.16	
58	0.5230	-14.6	0.2000	25.0	6.0	27.13	
65	0.7230	-11.2	0.2000	25.0	3.4	17.44	
71	0.9230	-7.7	0.2000	25.0	3.5	13.11	
77	1.1230	-5.4	0.2000	25.0	2.3	11.73	
83	1.3230	-3.2	0.2000	25.0	2.2	11.27	
88	1.5230	-0.5	0.2000	25.0	2.7	11.15	
94	1.7230	1.6	0.2000	25.0	2.1	10.90	
98	1.9230	3.9	0.2000	25.0	2.3	10.97	
103	2.1230	5.9	0.2000	25.0	2.0	10.69	
108	2.3230	8.0	0.2000	25.0	2.1	10.67	
113	2.5230	10.4	0.2000	25.0	2.4	11.05	
118	2.7230	12.5	0.2000	25.0	2.1	11.40	
123	2.9230	14.8	0.2000	25.0	2.3	12.08	
128	3.1230	17.4	0.2000	25.0	2.6	12.75	
133	3.3230	20.0	0.2000	25.0	2.6	13.66	
137	3.5230	22.9	0.2000	25.0	2.9	14.93	
142	3.7230	26.2	0.2000	25.0	3.3	16.23	
146	3.9230	29.7	0.2000	25.0	3.5	18.50	
150	4.1230	34.0	0.2000	25.0	4.3	23.29	
155	4.3230	39.3	0.2000	25.0	5.3	31.70	
159	4.5230	46.2	0.2000	25.0	6.9	50.82	
166	4.7230	56.6	0.2000	25.0	10.4	97.33	
174	4.8415	66.7	0.1185	25.0	10.1	182.81	
177	4.8980	73.6	0.0565	25.0	6.9	313.15	
180	4.9445	83.7	0.0465	25.0	10.1	481.28	

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188	4.9650	92.0	0.0205	25.0	8.3	528.89	EQP
191	4.9850	101.5	0.0200	25.0	9.5	572.59	
206	5.0050	125.0	0.0200	25.0	23.5	709.53	
	5.009554	131.3				711.33	
213	5.0250	152.6	0.0200	25.0	27.6	510.37	
216	5.0450	165.3	0.0200	25.0	12.7	541.61	
224	5.0720	173.2	0.0270	25.0	7.9	445.84	
227	5.1310	183.7	0.0590	25.0	10.5		
236	5.2050	189.2	0.0740	25.0	5.5		
239	5.3900	199.5	0.1850	25.0	10.3		
242	5.5775	205.4	0.1875	25.0	5.9		
246	5.7775	209.3	0.2000	25.0	3.9		

Sample Series

Name	Internal ID
Label 1	Label 2
Label 3	Label 4
Label 5	Label 6

Comment

Sample

Sample ID 1
Sample ID 2
Sample ID 3

Sample Parameters

Type	Name	Qualified name	Value
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Used Instruments

Name	T9_MSG_SA_002	Serial number	B551866924
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Responsible User

Time	Login name	Full name
05.02.2026 06:47:27	PredefinedUser	Administrator